

Today 9:33 AM

Flipkart Localstorage keys
CartActions:tooltipCount , NRBA_SESSION ,
UnivNav:Tooltip autosuggesthistory chatStartContext
isLoggedIn lastBTCallTime lastHeartbeatSentTime mypin
onReloadWebVitals pincodes

These look like **Flipkart web-app** `localStorage` **keys**. Their exact meaning can change with Flipkart releases, but based on the names, they likely fall into these categories:

Key	Likely purpose	Category	
<code>CartActions:tooltipCount</code>	Tracks how many times a cart-related tooltip/help UI has been displayed	UI state	
<code>NRBA_SESSION</code>	Likely session/analytics-related state used by an internal Flipkart module	Session / analytics	
<code>UnivNav:Tooltip</code>	Stores whether a tooltip in the universal navigation has been shown/dismissed	UI state	
<code>autosuggesthistory</code>	Stores recent search/autocomplete suggestions	Search	
<code>chatStartContext</code>	Context/state used when starting a customer-support/chat interaction	Support	
<code>isLoggedIn</code>	Client-side indication of login state	Authentication/UI	
<code>lastBTCallTime</code>	Timestamp associated with the last BT/background/backend-related call	Performance / networking	

Key	Likely purpose	Category
lastHeartbeatSentTime	Timestamp of the last heartbeat sent by the application	Session / connectivity
mypin	User's selected/detected PIN/postal code	Location
onReloadWebVitals	State/flag related to collecting Web Vitals after page reload	Performance monitoring
pincodes	Cached PIN-code/location information	Location

Important security observation

The interesting ones from a frontend/system-design perspective are:

1. isLoggedIn

This should **not be treated as authoritative authentication**.

```
localStorage
  ↓
isLoggedIn = true
  ↓
UI may show "logged in"
```



A malicious user can modify it:

```
</> JavaScript

localStorage.setItem('isLoggedIn', 'true');
```



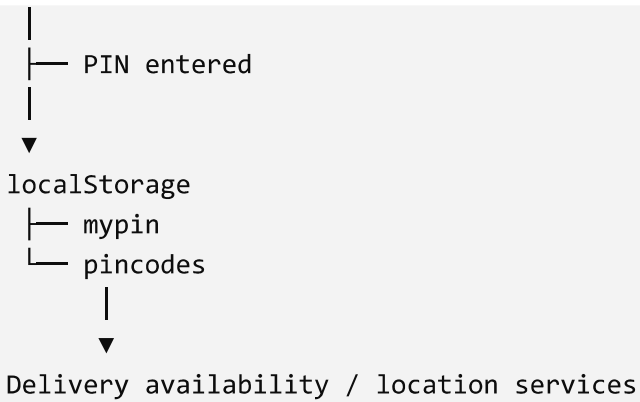
The server must still validate the actual authenticated session/token.

2. mypin / pincodes

These are probably convenience/cache data for location-aware functionality:

User

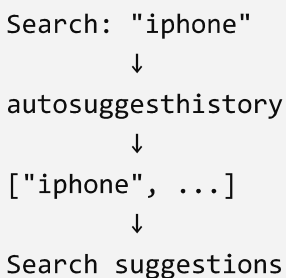




These aren't necessarily sensitive credentials, but they are still user-related data.

3. autosuggesthistory

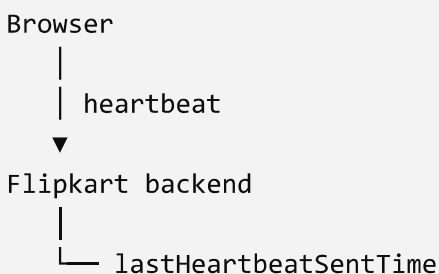
This is particularly interesting because it represents **user-generated search history**:



It improves UX but also means the browser can retain potentially sensitive searches.

4. lastHeartbeatSentTime

This suggests some client-side heartbeat mechanism:



It can be used to prevent excessive heartbeat requests or determine whether a heartbeat is due.

One important distinction

Seeing these keys **doesn't mean Flipkart stores authentication credentials in localStorage.**

For example:

```
localStorage
├── isLoggedIn
├── mypin
├── autosuggesthistory
└── UI / telemetry state
```



is very different from:

```
localStorage
├── access_token
├── refresh_token
└── session_secret
```



For a production application, I'd generally prefer:

```
Access Token → memory
Refresh Token → HttpOnly + Secure + SameSite cookie
Session      → HttpOnly + Secure cookie
UI preferences/cache → localStorage
```



That separation is especially important because **anything in localStorage is accessible to JavaScript running on the page**, so an XSS vulnerability can expose it.

If you're analyzing these keys for **Flipkart's architecture/interview/system-design purposes**, the most interesting next step is to classify them into **Authentication vs Session vs Cache vs Analytics vs UI State vs Performance**, and identify which should/shouldn't be in localStorage.



create an sample values of all keys

1. CartActions:tooltipCount - 6
2. NRBA_SESSION - {
"value": "2434ythh33434",

```
"inactiveAt": 2771784153688,  
"expiresAt": 2771896747064,  
"updatedAt": 2771645346450,  
"sessionReplay": 0,  
"sessionTraceMode": 0,  
"custom": {}
```

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Yes. If your goal is to **mock these Flipkart localStorage entries** for a frontend/system-design demo, you can create representative values like the following.

I'd recommend using **synthetic IDs, timestamps, URLs, and order numbers** rather than copying live session/order information.

Sample localStorage dataset

</> JavaScript



```
localStorage.setItem("CartActions:tooltipCount", "6");  
  
localStorage.setItem(  
  "NRBA_SESSION",  
  JSON.stringify(  
    value: "sample-session-2434ythh33434",  
    inactiveAt: 1786249513688,  
    expiresAt: 1786256747064,  
    updatedAt: 1786245346450,  
    sessionReplay: 0,  
    sessionTraceMode: 0,  
    custom: {}  
  ))  
);  
  
localStorage.setItem("UnivNav:Tooltip", "1");  
  
localStorage.setItem(  
  "autosuggesthistory",  
  JSON.stringify(  
    FLIPKART: [  
      {  
        action: {  
          action: null,  
          constraints: null,
```

```
customTrackingEvents: null,
fallback: null,
loginType: "LOGIN_NOT_REQUIRED",
nonWidgetizeRedirection: null,
omnitureData: null,
omnitureTracking: null,
originalUrl:
  "/search?q=wireless+headphones&sid=tyy,4mr,c3c&as=on&as-show=on",
params: {
  valid: true,
  query: "wireless+headphones",
  screenName: "productListView",
  store: "tyy/4mr/c3c"
},
requiredPermissionType: null,
screenType: "productListView",
storeValue: {},
subType: null,
tracking: {
  otracker: "AS_QueryStore_HistoryAutoSuggest_{POSITION}_0",
  otracker1: "AS_QueryStore_HistoryAutoSuggest_{POSITION}_0"
},
triggerExtraStandardEvents: null,
type: "NAVIGATION",
url:
  "/search?q=wireless+headphones&sid=tyy,4mr,c3c&as=on&as-show=on",
validationMeta: null,
widgetTracking: null,
widgetizedActions: null
},

tracking: {
  searchSource: "ASPrompt",
  otracker:
    "AS_QueryStore_OrganicAutoSuggest_{POSITION}_10_na_na_na",
  source: "ASPrompt",
  otracker1:
    "AS_QueryStore_OrganicAutoSuggest_{POSITION}_10_na_na_na",
  findingMethod: "search-autosuggest",
  requestId: "sample-request-001",
  suggestionId: "wireless headphones|Headphones",
  impressionId:
    "sample-request-001.wireless headphones|Headphones",
  contentType: "RECENT",
  position: "1"
},

value: {
  title: "wireless headphones",
```

```
    subTitle: "in Headphones",
    imageUrl:
      "https://example.com/flipkart-placeholder.png",
    query: "wireless head",
    template: "AUTOSUGGEST_QUERY_STORE_WIDGET",
    contentType: "HISTORY",
    backfill: "wireless headphones",
    navigationUrl:
      "/search?q=wireless+headphones&sid=tyy%2C4mr%2Cc3c&as=on&as-show=or",
    type: "QUERY_STORE"
  },

  marketplace: "FLIPKART",
  userTimeStamp: 1786246200382,
  version: "V4"
},

{
  value: {
    title: "kitchen storage containers",
    subTitle: "in Kitchen Storage",
    imageUrl:
      "https://example.com/flipkart-placeholder.png",
    query: "kitchen storage",
    template: "AUTOSUGGEST_QUERY_STORE_WIDGET",
    contentType: "HISTORY",
    backfill: "kitchen storage containers",
    navigationUrl:
      "/search?q=kitchen+storage+containers&as=on&as-show=on",
    type: "QUERY_STORE"
  },

  action: {
    url:
      "/search?q=kitchen%20storage%20containers&otracker=search",
    loginType: "LOGIN_NOT_REQUIRED",
    type: "mulitiwidget",
    params: {
      marketplace: "FLIPKART"
    },
    tracking: {
      otracker: "AS_Query_HistoryAutoSuggest_{POSITION}_0",
      otracker1: "AS_Query_HistoryAutoSuggest_{POSITION}_0"
    }
  },

  tracking: {
    searchSource: "ASPrompt",
    findingMethod: "search-autosuggest",
```

```
    requestId: "sample-request-002",
    suggestionId: "kitchen storage containers",
    impressionId: "sample-request-002.kitchen storage containers",
    contentType: "RECENT",
    position: "2"
  },

  marketplace: "FLIPKART",
  userTimeStamp: 1786245589960,
  version: "V4"
},

{
  value: {
    title: "running shoes",
    query: "running shoes",
    template: "AUTOSUGGEST_QUERY_WIDGET",
    backfill: "running shoes",
    contentType: "HISTORY",
    navigationUrl:
      "/search?q=running+shoes&otracker=search&marketplace=FLIPKART",
    type: "QUERY",
    imageUrl:
      "https://example.com/flipkart-placeholder.png"
  },

  action: {
    url:
      "/search?q=running%20shoes&otracker=search&marketplace=FLIPKART",
    loginType: "LOGIN_NOT_REQUIRED",
    type: "mulitiwidget",
    params: {
      marketplace: "FLIPKART"
    },
    tracking: {
      otracker: "AS_Query_HistoryAutoSuggest_{POSITION}_0",
      otracker1: "AS_Query_HistoryAutoSuggest_{POSITION}_0"
    }
  },

  marketplace: "FLIPKART",
  userTimeStamp: 1786244327426,
  version: "V4"
}
],

undefined: [
  {
    value: {
```



```

        title: "6 Seater Dining Table Cover",
        query: "6 Seater Dining Table Cover",
        template: "AUTOSUGGEST_QUERY_WIDGET",
        backfill: "6 Seater Dining Table Cover",
        contentType: "HISTORY",
        navigationUrl:
            "/search?q=6+Seater+Dining+Table+Cover&otracker=search",
        type: "QUERY",
        imageUrl:
            "https://example.com/flipkart-placeholder.png"
    },

    action: {
        url:
            "/search?q=6%20Seater%20Dining%20Table%20Cover&otracker=search",
        loginType: "LOGIN_NOT_REQUIRED",
        type: "mulitiwidget",
        params: {},
        tracking: {
            otracker: "AS_Query_HistoryAutoSuggest_{POSITION}_0",
            otracker1: "AS_Query_HistoryAutoSuggest_{POSITION}_0"
        }
    },

    userTimeStamp: 1786243231252,
    version: "V4"
}
]
})
);

localStorage.setItem(
    "chatStartContext",
    JSON.stringify({
        chatTenant: "flipkart_cx",
        chatType: "EHC",
        displayName: "Order OD123456789012345678",
        id: "OD123456789012345678",
        imageUrl: "https://example.com/product-placeholder.jpg",
        lang: "en",
        source: "order_desktop",
        type: "order",
        unitIds: ["123456789012345679"],
        salesExperience: "FLIPKART"
    })
);

localStorage.setItem("isLoggedIn", "true");

```

```

localStorage.setItem("lastBTCallTime", "1786249514341");

localStorage.setItem("lastHeartbeatSentTime", "1786249512005");

localStorage.setItem("mypin", "700140");

localStorage.setItem(
  "onReloadWebVitals",
  JSON.stringify({
    ev: [
      {
        name: "INP",
        value: 96
      },
      {
        name: "CLS",
        value: 0.024
      },
      {
        name: "LCP",
        value: 1820
      }
    ],
    nm: "gringotts-web-vitals",
    en: "AE",
    mt: {
      platform: "web",
      pageUri: "https://example.com/payments",
      userAgent:
        "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 Chrome/118.0.0.0",
      language: "en",
      appVersion: 0,
      reactVersion: "16.14.0",
      radio: "4g"
    }
  })
);

localStorage.setItem(
  "pincodes",
  JSON.stringify(["700140", "700091"])
);

```

Resulting localStorage structure

Conceptually, you'll have:

```
localStorage
├── CartActions:tooltipCount
│   └── "6"
├── NRBA_SESSION
│   └── { session metadata }
├── UnivNav:Tooltip
│   └── "1"
├── autosuggesthistory
│   └── {
│       FLIPKART: [...]
│       undefined: [...]
│   }
├── chatStartContext
│   └── { order/chat context }
├── isLoggedIn
│   └── "true"
├── lastBTCallTime
│   └── "1786249514341"
├── lastHeartbeatSentTime
│   └── "1786249512005"
├── mypin
│   └── "700140"
├── onReloadWebVitals
│   └── { INP, CLS, LCP, browser metadata }
└── pincodes
    └── ["700140", "700091"]
```

One important detail for your Angular application

Because `localStorage` stores strings only, objects/arrays need `JSON.stringify()` when writing and `JSON.parse()` when reading:

`</>` TypeScript

```
localStorage.setItem(
  'chatStartContext',
```

```
JSON.stringify(chatContext)
);

const chatContext = JSON.parse(
  localStorage.getItem('chatStartContext') ?? '{}'
);
```

For primitive values:

```
</> TypeScript

localStorage.setItem('isLoggedIn', 'true');
localStorage.setItem('mypin', '700140');

const isLoggedIn =
  localStorage.getItem('isLoggedIn') === 'true';

const pin =
  localStorage.getItem('mypin');
```

Note: `NRBA_SESSION` appears to contain session/telemetry state. Don't use a real session value from a live site in a mock application; use synthetic values like the example above.



create an image mentioning what keys we can kept inside
localstorage

What can we keep inside localStorage?

localStorage stores data as key-value pairs (string only).
Great for persisting **non-sensitive data** across browser sessions.

Good for client-side persistence.
Do NOT store sensitive or personal information.

Category	What to Store	Example Keys	Example Value (Sample)	Purpose / Use
User Preferences	Theme, language, layout, font size, UI preferences	theme language fontSize	"dark" "en" "16"	Remember user preferences across visits
Authentication State	Login status, tokens (if needed), user info (non-sensitive)	isLoggedIn authToken userInfo	"true" "eyJhbGciOiAi..." '{"id": "123", "name": "John"}'	Maintain login session or user state
Shopping / App Data	Cart items, recently viewed products, wishlist	cartItems recentlyViewed wishlist	'[{"id":1,"name":"Shoes","qty":1}]' '[1,2,3,4]' '[5,8,9]'	Improve shopping experience and restore state
Recent Activity / History	Search history, recent pages, form draft	searchHistory recentPages formDraft	'["shoes","tshirt","watch"]' '["/home","/products","/cart"]' '{"name": "John","email": "john@mail.com"}'	Provide quick access to recent activity
App Configuration / Settings	Feature flags, app settings, API base URL, environment	appConfig apiBaseUrl featureFlags	'{"itemsPerPage":20,"currency":"INR"}' "https://api.example.com" '{"newCheckout":true}'	Store application settings and configuration
Analytics / Performance	Web vitals, analytics preferences, last visited time	webVitals lastVisitedAt analyticsOptIn	'{"INP":96,"CLS":0.024,"LCP":1820}' "17862495514341" "true"	Track performance metrics or analytics preferences
Location / Address	Saved addresses, PIN codes, delivery locations	pinCodes selectedAddress	'["700140","700091"]' '{"name":"Home","pin":"700140","city":"Kolkata"}'	Faster checkout and location-based features
Chat / Support	Chat session context, last conversation info	chatStartContext lastChatId	'{"chatTenant":"flipkart_cx","type":"order","id":"00123"}' "chat_123456"	Restore chat context and improve support flow

Remember: localStorage capacity is ~5-10MB (varies by browser).

Best for: Non-sensitive, persistent, small to medium data.

Avoid: Passwords, OTPs, card details, tokens with high security risk.

Would you like a ↩️ **dark-tech version** or a ↩️ **minimal interview/LinkedIn carousel version** of this localStorage infographic?



ChatGPT can make mistakes. Check important info.